Aman Syed

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PERSONAL PROFILE

Associate Data Scientist and AI/ML Engineer with 2+ years delivering production-grade LLM systems and ML data pipelines on AWS. Cut manual analysis time by 95% through multi-agent orchestration, built GPU-accelerated inference pipelines processing millions of daily articles, and deployed RAG architectures with vector database search at sub-500ms latency. Works across the full ML lifecycle — context engineering, prompt engineering, model evaluation, inference optimisation, and production serving — using OpenAI, Gemini, and Claude. **Core stack: Python, SQL, PyTorch, FastAPI, Docker, PostgreSQL, SageMaker**.

EXPERIENCE

Associate Data Scientist *Oct'24 – Present*
Penta Group, London

- **Developed conversational RAG system** with SSE streaming, message history context engineering, & adaptive context windowing based on entity/keyword anchors — achieving 62% content reduction while maintaining answer quality; integrated adaptive LLM routing based on query complexity, query enhancement pipeline, NMF topic diversification, & structured inline citations; reduced LLM costs by 60%.
- **Engineered hybrid search engine** combining vector (pgvector HNSW), full-text (tsvector), and NER entity search with reciprocal rank fusion across 6 dynamic weight tiers; implemented two-tier cache (exact hash + semantic similarity) and per-user token bucket rate limiting.
- **Built MCP (Model Context Protocol) server** exposing 5 AI-client tools — hybrid search, vector search, full-text search, entity search, and RAG Q&A — over SSE with JWT authentication; enables external AI agents to query the database article knowledge base with cited answers via stateless, config-driven endpoints.
- **Scaled GPU-accelerated NLP data pipeline** to process 2–3 million daily articles via SQS-driven workers running GLiNER NER and EmbeddingGemma-300m on NVIDIA GPUs; reduced redundant inference through 30-day SHA256 content deduplication and deployed pgvector HNSW-indexed vector database search with daily-partitioned PostgreSQL tables on AWS ECS.
- **Deployed RAG-powered AI Narrative engine** generating sentiment-driven narratives for automated PowerPoint presentations across 16 client-facing modules, used in daily briefings; engineered async retrieval using K-Means/HDBSCAN clustering with MMR diversity, primary/fallback LLM architecture, and Parquet-based embedding caching via FastAPI endpoints.
- **Reduced manual transcript analysis** time by 95% (20 hrs to under 5 min) by orchestrating a multi-agent NLP platform processing text, audio and video inputs via Rev AI speech-to-text, 3-LLM consensus speaker extraction (Gemini/OpenAI/Claude), and a 3-stage quote extraction pipeline — relevance filtering, subquote precision extraction, and quality evaluation with composite re-ranking — achieving 92% precision and reducing manual review for analysts.
- **Improved sentiment model** accuracy to 71% weighted F1 on 1,000 human-labeled articles by evaluating 33+ prompt configurations across 6 LLMs (Gemini, Claude, GPT) and benchmarking Llama 3.1/3.2 fine-tuning vs zero-shot approaches; reduced per-inference costs via output token capping and model selection.
- **Automated client newsletter curation** by engineering a summarization pipeline processing 2,000+ articles per portfolio using multi-provider LLM architecture (GPT-4o/5, Gemini 2.5, Claude 4) with prompt optimisation framework and multi-model A/B testing.
- **Productionized AI scraper** adopted by engineering team for structured data ingestion; evaluated CSS selector generation across LLMs with 9-step HTML cleaning (70–80% size reduction) and 4-tier validation.
- **Achieved 3x throughput** and sub-500ms latency by deploying BERT-based models as serverless SageMaker inference endpoints with dynamic batching, and built dual LDA/NMF topic extraction pipeline with automatic model selection and interactive network/treemap visualisations.
- **Delivered 5x faster** file ingestion across all internal tools via chunked-transfer uploads over parallel WebSockets; migrated hardcoded credentials to PostgreSQL and AWS SSM across dev/pilot/prod environments.

Financial Research Assistant *Sep'24 – Oct'24*
The University of Edinburgh – Business School, Edinburgh

- **Developed Python data pipeline using Refinitiv API** to collect financial data across 500+ U.S. companies spanning previous 3 fiscal years for environmental research, automating data extraction that previously required manual retrieval.
- **Compiled earnings call transcripts and presentation briefs** from Refinitiv Workspace, building a structured dataset for text and sentiment analysis of corporate environmental disclosures.

Data Engineer Intern *Dec'22 – Aug'23*
C.H. Robinson, Mumbai

- **Designed & implemented a serverless** data pipeline on AWS (Lambda, S3, API Gateway), processing 100K daily API records to improve data availability and system reliability.
- **Automated workflow** processes using **Selenium & Python scripts**, saving 90+ monthly engineering hours and doubling data capture capacity.
- **Revamped** logistics reporting by **automating daily client** report generation with Python, increasing output to 20+ reports daily and boosting client satisfaction.
- **Optimized 200+ AWS Lambda** functions through SQL query tuning and Python code refactoring, improving data processing efficiency by 40%.

PROJECTS

Finance Flow (frontend · backend) *2024 – Present*
React 19, TypeScript, Tailwind CSS, FastAPI, Google Sheets API, Render

- Built full-stack personal finance suite: React 19 dashboard with analytics, budgets, dark mode, and Google Sheets sync; Python FastAPI backend using the Strategy Pattern to auto-detect and normalise CSVs from 6 UK banks (Amex, Revolut, Monzo, Starling, BoS) into a unified schema with Frankfurter exchange rate proxy. Deployed frontend on GitHub Pages, backend on Render.

Knock Knock Physio (source) *2024*
React, TypeScript, Vite, GitHub Pages

- Launched a production business website for a mobile physiotherapy practice — online booking, WhatsApp integration, pricing plans, testimonials, and protected admin dashboard.

TECHNICAL SKILLS

- **Languages:** Python, SQL, TypeScript, JavaScript, R, Java
- **LLMs & Gen AI:** OpenAI GPT, Google Gemini API, Anthropic Claude, RAG, LLM Fine-tuning, Prompt Engineering, HuggingFace Transformers, Multi-agent Orchestration
- **ML/NLP:** PyTorch, SpaCy, BERT, GLiNER, Gensim, Scikit-learn, Pandas, NumPy, LDA/NMF, Sentence Embeddings
- **Frontend & Web:** React 19, Vite, TailwindCSS, Recharts, Framer Motion
- **Backend, DBs & APIs:** FastAPI, Flask, Pydantic, PostgreSQL (pgvector), MySQL, MongoDB, REST APIs
- **Cloud (AWS):** SageMaker, ECS, Lambda, S3, ASG, SQS, API Gateway, CloudWatch, SSM Parameter Store, ECR
- **MLOps & DevOps:** Docker, Git, CI/CD, GitHub Actions, GitHub Pages, Render, Inference Optimisation, Agile/Scrum
- **Other:** Selenium, Refinitiv Workspace, Tesseract, Postman, Matplotlib, Plotly, Power BI

EDUCATION

MSc Statistics with Data Science *Pass*
The University of Edinburgh, Edinburgh *Sep'23 – Aug'24*

BEng Computer Engineering *9.83/10 CGPA*
University of Mumbai, Mumbai *Aug'19 – Jun'23*

CERTIFICATIONS

- Architecting with Google Compute Engine Specialization — Google Cloud
- Machine Learning Foundations: A Case Study Approach — University of Washington